

EXTRACTING DATA

1. Create table and insert values

Firstly, Creating the table student and inserted the values to perform the operations such as SELECT, WHERE, ORDER BY, LIMIT, LIKE, BETWEEN & AND...

The screenshot shows the SQL Server Enterprise Manager interface. On the left, the Object Explorer displays the database structure for 'localhost (SQL Server 16.0.1000.6 - SIVU\Krishy)'. The 'Tables' folder is expanded, showing the 'student' table. On the right, the SQL Query window displays the following SQL script:

```
1 CREATE TABLE student (  
2     student_id INT PRIMARY KEY,  
3     name VARCHAR(100),  
4     age INT,  
5     gender VARCHAR(10),  
6     CGPA DECIMAL(5,2),  
7     city VARCHAR(100),  
8     enrollment_date DATE  
9 );  
10 INSERT INTO student (student_id, name, age, gender, CGPA, city, enrollment_date)  
11 VALUES  
12 (1, 'Aarav Kumar', 17, 'Male', 88.50, 'Chennai', '2023-06-15'),  
13 (2, 'Meera Iyer', 18, 'Female', 92.00, 'Coimbatore', '2022-08-20'),  
14 (3, 'Rohan Das', 16, 'Male', 76.25, 'Chennai', '2023-01-10'),  
15 (4, 'Divya Raj', 17, 'Female', 81.75, 'Madurai', '2023-03-05');  
16 select * from student
```

Below the script, the 'Results' tab shows the output of the query, displaying a table with 7 columns: student_id, name, age, gender, CGPA, city, and enrollment_date. The data is as follows:

student_id	name	age	gender	CGPA	city	enrollment_date
1	Aarav Kumar	17	Male	88.50	Chennai	2023-06-15
2	Meera Iyer	18	Female	92.00	Coimbatore	2022-08-20
3	Rohan Das	16	Male	76.25	Chennai	2023-01-10
4	Divya Raj	17	Female	81.75	Madurai	2023-03-05

2. Select Statement

Filtering the values using various conditions by using select query with orderby

The screenshot shows the SQL Query window with the following SQL script:

```
1 select * from student  
2  
3 --select top 5 results  
4 select top 5 * from student order by CGPA
```

Below the script, the 'Results' tab shows the output of the query, displaying a table with 7 columns: student_id, name, age, gender, CGPA, city, and enrollment_date. The data is as follows:

student_id	name	age	gender	CGPA	city	enrollment_date
1	Aarav Kumar	17	Male	88.50	Chennai	2023-06-15
2	Meera Iyer	18	Female	92.00	Coimbatore	2022-08-20
3	Rohan Das	16	Male	76.25	Chennai	2023-01-10
4	Divya Raj	17	Female	81.75	Madurai	2023-03-05

SQLQuery2.sql... \Krishy (60))* SQLQuery1.sql ...U\Krishy (63))*

```
5
6  --list names of top 3 students
7  select top(3) name from student order by CGPA
```

100 % 6 0

Results Messages

	name
1	Rohan Das
2	Divya Raj
3	Aarav Kumar

```
17
18  --top 5 students from Chennai scoring between 80 and 95.
19  SELECT * FROM student
20  WHERE CGPA BETWEEN 80 AND 95
21        AND city LIKE 'Chennai%'
22  ORDER BY CGPA DESC
23
```

100 % 2 0

Results Messages

	student_id	name	age	gender	CGPA	city	enrollment_date
1	1	Aarav Kumar	17	Male	88.50	Chennai	2023-06-15

3. Where Condition

Here we see how Where clause can be used with different functions...

SQLQuery1.sql... \Krishy (52))* a4j4sqj1..sql -...VU\Krishy (101)) cavxfyuf..sql -...IVU\Krishy (99))

```
1 --students who are exactly 18 years old
2 SELECT * FROM student WHERE age = 18;
3
4 --male students who has above 80% CGPA
5 SELECT * FROM student WHERE gender = 'Male' AND CGPA > 80;
6
7 --students who came from Coimbatore or Madurai
8 SELECT * FROM student WHERE city = 'Coimbatore' or city = 'Madurai'
9
10
```

100 % No issues found

Results Messages

	student_id	name	age	gender	CGPA	city	enrollment_date
1	2	Meera Iyer	18	Female	92.00	Coimbatore	2022-08-20

	student_id	name	age	gender	CGPA	city	enrollment_date
1	1	Aarav Kumar	17	Male	88.50	Chennai	2023-06-15

	student_id	name	age	gender	CGPA	city	enrollment_date
1	2	Meera Iyer	18	Female	92.00	Coimbatore	2022-08-20
2	4	Divya Raj	17	Female	81.75	Madurai	2023-03-05

SQLQuery1.sql... \Krishy (52))* a4j4sqj1..sql -...VU\Krishy (101)) cavxfyuf..sql -...IVU\Krishy (99)) dwjtgpp0..sql ...VU\Krishy (98))

```
9
10 --students whose name ends with letter r
11 SELECT * FROM student WHERE name LIKE '%r';
12
13 --students who joined in the year of 2023
14 SELECT * FROM student WHERE enrollment_date BETWEEN '2023-01-01' AND '2023-12-31';
15
16
17
18
```

100 % 1 0

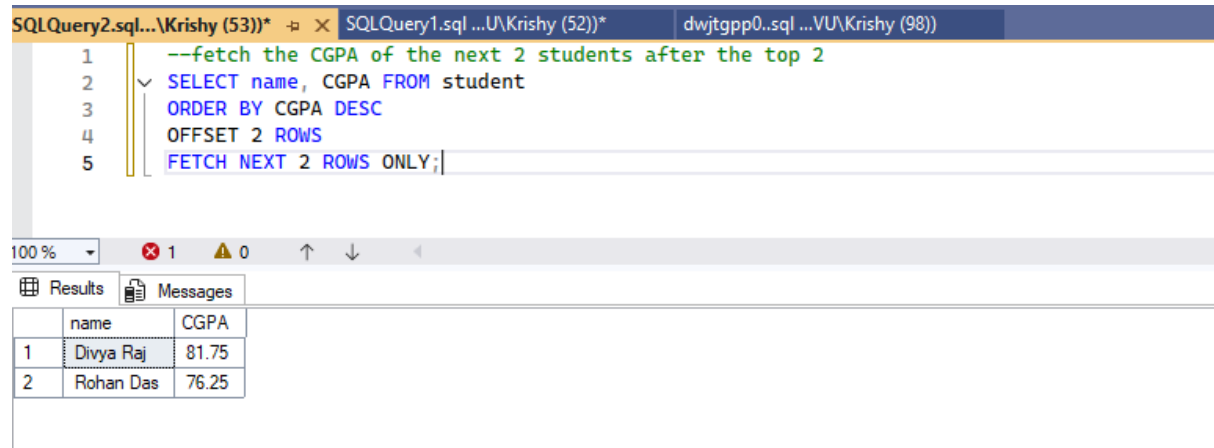
Results Messages

	student_id	name	age	gender	CGPA	city	enrollment_date
1	1	Aarav Kumar	17	Male	88.50	Chennai	2023-06-15
2	2	Meera Iyer	18	Female	92.00	Coimbatore	2022-08-20

	student_id	name	age	gender	CGPA	city	enrollment_date
1	1	Aarav Kumar	17	Male	88.50	Chennai	2023-06-15
2	3	Rohan Das	16	Male	76.25	Chennai	2023-01-10
3	4	Divya Raj	17	Female	81.75	Madurai	2023-03-05

4. Limit with orderby

SQL Server does not support LIMIT. Instead, it uses the TOP keyword or OFFSET and FETCH.



The screenshot shows a SQL Server Enterprise Manager window with a query editor and a results pane. The query editor contains the following SQL code:

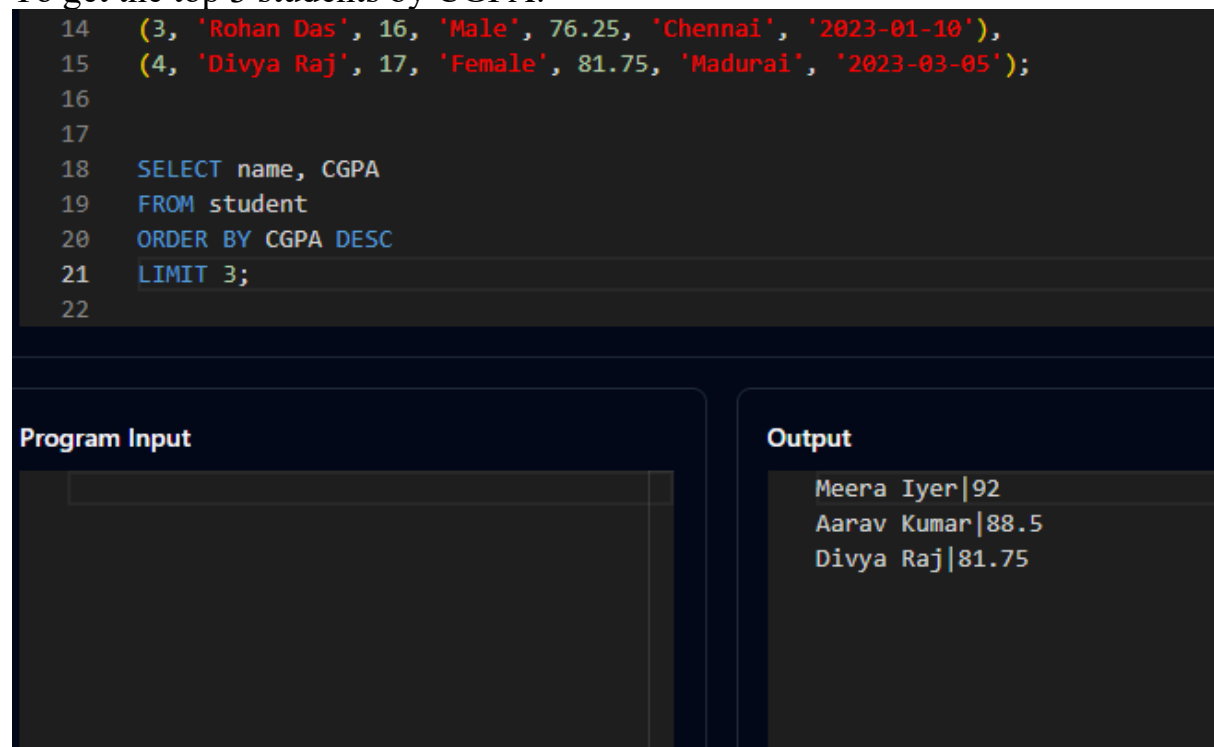
```
1  --fetch the CGPA of the next 2 students after the top 2
2  SELECT name, CGPA FROM student
3  ORDER BY CGPA DESC
4  OFFSET 2 ROWS
5  FETCH NEXT 2 ROWS ONLY;
```

The results pane shows the following table:

	name	CGPA
1	Divya Raj	81.75
2	Rohan Das	76.25

Limit will be supported by MySQL, PostgreSQL, SQLite

To get the top 3 students by CGPA:



The screenshot shows a code editor with the following SQL code:

```
14  (3, 'Rohan Das', 16, 'Male', 76.25, 'Chennai', '2023-01-10'),
15  (4, 'Divya Raj', 17, 'Female', 81.75, 'Madurai', '2023-03-05');
16
17
18  SELECT name, CGPA
19  FROM student
20  ORDER BY CGPA DESC
21  LIMIT 3;
22
```

The output pane shows the following results:

```
Meera Iyer|92
Aarav Kumar|88.5
Divya Raj|81.75
```

To get the next 3 after top 3 students by CGPA:

```
14 (3, 'Rohan Das', 16, 'Male', 76.25, 'Chennai', '2023-01-10'),
15 (4, 'Divya Raj', 17, 'Female', 81.75, 'Madurai', '2023-03-05');
16
17
18 SELECT name, CGPA
19 FROM student
20 ORDER BY CGPA DESC
21 LIMIT 3 OFFSET 3;
22
```

Program Input

Output
Rohan Das | 76.25

5. Distinct

Give unique entries of the record without any duplicates.

```
SQLQuery4.sql...\\Krishy (63))*  X SQLQuery3.sql ...\\Krishy (61))*  SQLQuery2.sql ...\\Krishy (53))*  SQLQuery1.sql .
1  --list all cities where students are from
2  SELECT DISTINCT city
3  FROM student;
```

100 % No issues found

Results Messages

	city
1	Chennai
2	Coimbatore
3	Madurai

THANK YOU!